

translate the latter feeling into energy through which they can continue to improve and reach their potential. It is the physician's responsibility to pursue new knowledge continually by reading, attending conferences and courses, and consulting colleagues and the Internet. This is often a difficult task for a busy practitioner; however, a commitment to continued learning is an integral part of being a physician and must be given the highest priority.

The Physician as Citizen Being a physician is a privilege. The capacity to apply one's skills for the benefit of fellow human beings is a noble calling. The physician–patient relationship is inherently unbalanced in the distribution of power. In light of their influence, physicians must always be aware of the potential impact of what they do and say and must always strive to strip away individual biases and preferences to find what is best for their patients. To the extent possible, physicians should also act within their communities to promote health and alleviate suffering. Meeting these goals begins by setting a healthy example and continues in taking action to deliver needed care even when personal financial compensation may not be available.

Research, Teaching, and the Practice of Medicine The word *doctor* is derived from the Latin *docere*, “to teach.” As teachers, physicians should share information and medical knowledge with colleagues, students of medicine and related professions, and their patients. The practice of medicine is dependent on the sum total of medical knowledge, which in turn is based on an unending chain of scientific discovery, clinical observation, analysis, and interpretation. Advances in medicine depend on the acquisition of new information through research, and improved medical care requires the transmission of that information. As part of their broader societal responsibilities, physicians should encourage patients to participate in ethical and properly approved clinical investigations if these studies do not impose undue hazard, discomfort, or inconvenience. Physicians engaged in clinical research must be alert to potential conflicts of interest between their research goals and their obligations to individual patients. The best interests of the patient must always take priority.

To wrest from nature the secrets which have perplexed philosophers in all ages, to track to their sources the causes of disease, to correlate the vast stores of knowledge, that they may be quickly available for the prevention and cure of disease—these are our ambitions.

—William Osler, 1849–1919

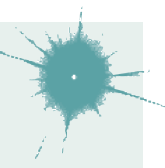
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2

Promoting Good Health

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GOALS AND APPROACHES TO PREVENTION

Prevention of acute and chronic diseases before their onset has been recognized as one of the hallmarks of excellent medical practice for centuries and is now used as a metric for highly functioning health care systems. The ultimate goal of preventive strategies is to avoid premature death. However, as longevity has increased dramatically worldwide over the last century (largely as a result of public health practices), increasing emphasis is placed on prevention for the purpose of preserving quality of life and extending the health span, not just the life span. Given that all patients will eventually die, the goal of prevention ultimately becomes compression of morbidity toward the end of the life span; that is, reduction of the amount of burden and time spent with disease prior to dying. As shown in Fig. 2-1, normative aging tends to involve a steady decline in the stock of health, with accelerating decline over time. Successful prevention offers the opportunity both to extend life and to extend healthy life, thus “squaring the curve” of health loss during aging.

Prevention strategies have been characterized as tertiary, secondary, primary, and primordial. *Tertiary prevention* requires rapid action to prevent imminent death and preserve organ function in the setting of acute illness, such as through thrombolysis or thrombectomy in acute ischemic stroke. *Secondary prevention* strategies focus on avoiding the recurrence of disease and death in an individual who is already affected. For example, tamoxifen is recommended for women with surgically treated early-stage, estrogen receptor–positive breast cancer, because it reduces the risk of recurrent breast cancer (including in the contralateral breast) and death. *Primary prevention* attempts to reduce the risk of incident disease among individuals with one or more risk factors. Treatment of elevated blood pressure in individuals who have not yet experienced cardiovascular disease represents one example of primary prevention that has proven effective in reducing the incidence of stroke, heart failure, and coronary heart disease.

Primordial prevention is a more recent concept (first introduced in 1979) that focuses on prevention of the development of *risk factors* for disease, not just prevention of disease. Primordial prevention strategies emphasize upstream determinants of risk for chronic diseases, such as eating patterns, physical activity, and environmental and social determinants of health. It therefore encompasses medical treatment strategies for some individuals as well as a strong reliance on public health and social policy. It is increasingly clear that primordial prevention represents the ultimate means for reducing the burden of chronic

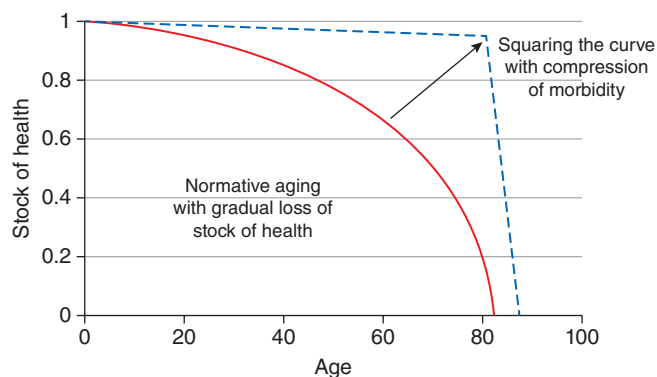


FIGURE 2-1 Loss of health with aging. Representation of normative aging with loss of the full stock of health with which individuals are born (indicating gain of morbidity), contrasted with a squared curve with greater longevity and fuller stock of health (less morbidity) until shortly before death. The “squared curve” represents the likely ideal situation for most patients.

diseases of aging. Once risk factors develop, it is difficult to restore risk to the low level of someone who never developed the risk factor. The time spent with adverse levels of the risk factor often causes irreversible damage that precludes complete restoration of low risk. For example, individuals with hypertension who are treated back to optimal levels (<120/<80 mmHg) do have a lower risk compared with untreated patients with hypertension, but they still have twice the risk of cardiovascular events as those who maintained optimal blood pressure without medications. Patients with elevated blood pressure that is subsequently treated have greater left ventricular mass index, worse renal function, and more evidence of atherosclerosis and other target organ damage as a result of the time spent with elevated blood pressure; such damage cannot be fully reversed despite efficacious therapy with antihypertensive medications. Conversely, as described below in greater detail, individuals who maintain optimal levels of all major cardiovascular risk factors into middle age through primordial prevention essentially abolish their lifetime risk of developing cardiovascular disease while also living substantially longer and having a lower burden and later onset of other chronic comorbid illnesses (compression of morbidity).

Prevention strategies should be distinguished from disease-screening strategies. Screening attempts to detect evidence of disease at its earliest stages, when treatment is likely to be more efficacious than for advanced disease (Chap. 6). Screening can be performed in service of prevention, especially if it aids in identifying preclinical markers, such as dyslipidemia or hyperglycemia, associated with elevated disease risk.

■ HEALTH PROMOTION

In recent decades, medical practice has increasingly focused on clinical and public health approaches to promote health and not just prevent disease. Prevention of disease is a worthy individual and societal goal in and of itself, but it does not necessarily guarantee health. Health is a broader construct encompassing more than just absence of disease. It includes biologic, physiologic, and psychological domains (among others) in a continuum, rather than occurring as a dichotomous trait. Health is therefore somewhat subjective, but attempts have been made to use more objective criteria to define health in order to raise awareness, prevent disease, and promote healthy longevity.

For example, in 2010, the American Heart Association (AHA) formally defined a new construct of “cardiovascular health” based on evidence of associations with longevity, disease avoidance, healthy longevity, and quality of life. The concept was updated and expanded in 2022 to encompass eight metrics that help define an individual’s or population’s current health status. The eight metrics (termed Life’s Essential 8™) include diet, physical activity, sleep, nicotine exposure, body mass index, blood lipids, glycemia, and blood pressure. Each metric is scored on a scale of 0 to 100 points (higher is better), and overall health is measured as the average of the scores (also from 0 to 100). Higher cardiovascular health scores at all ages have been associated with greater longevity, lower incidence of cardiovascular disease, lower incidence of other chronic diseases of aging (including dementia, cancer, diabetes, and more), compression of morbidity, greater quality of life, and lower health care costs, achieving both individual and societal goals for healthy aging and further establishing the critical importance of primordial prevention and health promotion.

Focusing on health promotion, rather than just disease prevention, may also provide greater motivation for patients to pursue lifestyle changes or adhere to clinician recommendations. Extensive literature suggests that providing patients solely with information regarding disease risk, or risk reduction with treatment, is unlikely to motivate desired behavior change. Empowering patients with strategies to achieve positive health goals after discussing risks can provide more effective adherence and better long-term outcomes. In the case of smoking cessation, enumerating only the risks of smoking can lead to patient inertia and therapeutic nihilism and has proven to be an ineffective approach, whereas strategies that incorporate positive health messaging, support, and feedback, with appropriate use of evidence-based therapies, have proven far more effective.

■ PRIORITIZING PREVENTION STRATEGIES

In secondary prevention, the patient already has manifest clinical disease and is therefore at high risk for progression. The approach should be to work with the patient to implement all evidence-based strategies that will help to prevent recurrence or progression. This will typically include drug therapy as well as therapeutic lifestyle changes to control ongoing risk factors that may have caused disease in the first place. Juggling priorities can be difficult, and barriers to implementation are many, including costs, time, patient health literacy, and patient and caregiver capacity to organize the regimen. Addressing these potential barriers with the patient can help to forge a therapeutic bond and may improve adherence; ignoring them will likely lead to therapeutic failure. Numerous studies demonstrate that, even in high-functioning health systems, only ~50% of patients are taking recommended, evidence-based secondary prevention medications, such as statins, by 1 year after a myocardial infarction.

In patients who are eligible for primary prevention strategies, it is important to frame the discussion around the overall evidence base as well as an individual patient’s likelihood of benefit from a given preventive intervention. A first step is to understand the patient’s estimated absolute risk for disease in the foreseeable future or during their remaining life span. However, absolute risk estimation and presentation of those risks are generally insufficient to motivate behavior change. It is critical to assess the patient’s understanding and tolerance of the risk, their readiness to implement lifestyle changes or adhere to drug therapy, and their overall preferences regarding use of drug therapy to prevent an event (e.g., cancer, myocardial infarction, stroke). The clinician can help the patient by informing them of the risks for disease and potential for absolute benefits (and harms) from the available evidence-based choices. This may take more than one conversation, but given that diseases, such as cancer and cardiovascular disease, are the leading causes of premature death and disability, the time is well spent.

Partnering with the patient through motivational interviewing may assist in the process of selecting initial approaches to prevention. Selecting an area that the patient feels they are ready to change can lead to better adherence and greater achievement of success in the short and longer term. If the patient is uncertain what course to choose, prudence would dictate focusing on control of risk factors that may lead to the most rapid reduction in risk for acute events. For example, blood pressure is both a chronic risk factor and an acute trigger for cardiovascular events. Thus, if a patient has both significant elevations in blood pressure and dyslipidemia, it would be appropriate to focus initial efforts on blood pressure control. Likewise, a focus on smoking cessation can lead to more rapid reductions in risk for acute events than some other lifestyle interventions.

■ PREVENTION AND HEALTH PROMOTION ACROSS THE LIFE COURSE

Periodic Health Evaluations The “routine annual physical” has in many ways become an expected part of the patient-physician relationship in primary care practice. However, evidence for the efficacy of the periodic health evaluation in asymptomatic adults unselected for risk factors or disease is mixed and depends on the study design and outcome. Systematic reviews and meta-analyses of published randomized trials have consistently observed lack of benefit (and also lack of harm) in terms of total mortality in association with periodic health evaluations. Data are more heterogeneous but overall suggest no benefit for cancer- or cardiovascular-specific mortality, with the potential for either benefit or harm depending on number of evaluations and patient-level factors. Well-designed studies on nonfatal clinical events and morbidity have been sparsely reported, but there appear to be no large effects.

Periodic health evaluations do appear to lead to greater diagnosis of certain conditions such as hypertension and dyslipidemia, as expected. Likewise, periodic health examinations also improve the delivery of recommended preventive services, such as gynecologic examinations and Papanicolaou smears, fecal occult blood testing, and cholesterol

screening. The benefits and risks associated with screening tests are discussed in detail in [Chap. 6](#). Risks of routine evaluations include inappropriate testing or overtesting, or false-positive findings that require follow-up and induce patients to worry. Periodic health examinations appear to be associated with less patient worry. On balance, given the lack of convincing evidence of harm and the potential for better delivery of appropriate screening, counseling, and preventive services, periodic health evaluations appear reasonable for general populations at average risk for chronic conditions.

It is important to note that routine annual comprehensive *physical* examinations of asymptomatic adult patients have very low yield and may take an inordinate amount of time in a wellness visit. Such time may be better spent on assessing and counseling the patient on other aspects of their health, as discussed below. Evidence-based components that should be included in periodic evaluations focused on health and prevention include a number of age-appropriate screening tests for chronic disease and risk factors, preventive interventions including immunizations and chemoprevention for at-risk individuals, and preventive counseling. The U.S. Preventive Services Task Force publishes its *Guide to Clinical Preventive Services*, which contains evidence-based recommendations from the Task Force on preventive services for which there is a high degree of certainty that the service provides at least moderate net clinical benefit (i.e., benefits outweigh harms significantly and to a reasonable magnitude).

Healthy Behaviors and Lifestyles Owing to the heterogeneity of study designs and the diverse nature of lifestyle interventions studied, many clinicians are uncertain as to how to deliver advice regarding healthy behaviors and lifestyles. Nevertheless, adverse behaviors and lifestyles contribute to >75% of premature, preventable deaths and disability. Estimates from the U.S. National Health and Nutrition Examination Survey indicate that fewer than 1% of Americans achieve an optimal heart-healthy eating pattern. Thus, whereas there are many

demands on time during a typical patient-clinician encounter, few things may have more impact on longevity, health, and quality of life for asymptomatic patients than an efficient approach to assessing, documenting, and improving patients' health behaviors. Indeed, the mere act of assessing health behaviors has been shown to affect patients' health behaviors. Facility with tools for assessment of lifestyle and with strategies for counseling are therefore of paramount importance.

Healthy Eating Patterns (see Chap. 343) Despite the existence of numerous “fad” diets and seemingly inconsistent recommendations on dietary composition, there is remarkable agreement about what should constitute a healthy eating pattern for the broad population to avoid nutritional deficits (i.e., vitamin deficiency) and excesses (i.e., excessive caloric intake) and to maximize potential health ([Table 2-1](#)). Optimal eating patterns consist of whole fruits and vegetables, whole grains, lean proteins, and healthy oils, and allow for nonfat or low-fat dairy intake. They tend to exclude frequent ingestion of foods high in refined sugars and starches, saturated fat, and sodium. Since sodium and refined sugars and starches are the hallmark of much of the processed/packaged food supply, a simple rule of thumb is to provide or cook the majority of one's own meals starting from whole foods and emphasizing fruits and vegetables. Likewise, foods prepared outside of the home tend to have higher fat and sodium content, so special attention to menu choices focused on fruits, vegetables, lean proteins, and whole grains, while minimizing sauces and dressings, can help most individuals follow healthier eating patterns when eating food prepared outside the home. In all cases, sugar-sweetened beverages and non-nutritious snack foods should be minimized. If snacks are included, small amounts of healthy nuts and seeds or more fruits and vegetables should be encouraged.

Specific conditions and diseases, such as diabetes or hypertension, other metabolic disorders, allergies, and gastrointestinal disorders, may require tailored approaches to diet. In counseling most patients,

TABLE 2-1 Guidelines and Key Recommendations from the <i>Dietary Guidelines for Americans, 2020–2025</i>	
GUIDELINES	KEY RECOMMENDATIONS
1. Follow a healthy dietary pattern at every life stage. For the first 6 months of life, infants should be exclusively fed human milk, or iron-fortified formula if human milk is unavailable. From 6 to 12 months, infants should be introduced to a variety of complementary nutrient-dense foods. From 12 months to older adulthood, the dietary pattern should meet nutrient needs, help achieve a healthy body weight, and reduce the risk of chronic disease. 2. Customize and enjoy nutrient-dense food and beverage choices to reflect personal preferences, cultural traditions, and budgetary considerations. The Dietary Guidelines provide a framework of several dietary patterns intended to be customized to individual needs and preferences, as well as the foodways of the diverse cultures in the United States. 3. Focus on meeting food group needs with nutrient-dense foods and beverages and stay within calorie limits. Nutrient-dense foods provide vitamins, minerals, and other health-promoting components and have no or little added sugars, saturated fat, and sodium. A healthy dietary pattern consists of nutrient-dense forms of foods and beverages across all food groups, in recommended amounts, and within calorie limits. 4. Limit foods and beverages higher in added sugars, saturated fat, and sodium, and limit alcoholic beverages. At every life stage, meeting food group recommendations, even with nutrient-dense choices, fulfills most of a person's daily calorie needs and sodium limits, with little room for extra added sugars, saturated fat, or sodium, or for alcoholic beverages.	The Dietary Guidelines' Key Recommendations for healthy eating patterns should be applied in their entirety, given the interconnected relationship that each dietary component can have with others. They are also intended as a framework to accommodate personal preferences, cultural traditions, and budgetary considerations. Focus on meeting food group needs with nutrient-dense foods and beverages, and stay within calorie limits to achieve a healthy weight and reduce the risk of chronic disease. The core elements that make up a healthy dietary pattern include: • Vegetables of all types—dark green; red and orange; beans, peas, and lentils; starchy; and other vegetables • Fruits, especially whole fruit • Grains, at least half of which are whole grain • Dairy, including fat-free or low-fat milk, yogurt, and cheese, and/or lactose-free versions and fortified soy beverages and yogurt as alternatives • Protein foods, including lean meats, poultry, and eggs; seafood; beans, peas, and lentils; and nuts, seeds, and soy products • Oils, including vegetable oils and oils in food, such as seafood and nuts A healthy eating pattern limits: • Added sugars—Less than 10% of calories per day starting at age 2. Avoid foods and beverages with added sugars for those younger than age 2. • Saturated fat—Less than 10% of calories per day starting at age 2. • Sodium—Less than 2300 mg/d—and even less for children younger than age 14. • Alcoholic beverages—Adults of legal drinking age can choose not to drink or to drink in moderation by limiting intake to 2 drinks or less in a day for men and 1 drink or less in a day for women, when alcohol is consumed. Drinking less is better for health than drinking more. There are some adults who should not drink alcohol, such as women who are pregnant. Meet the U.S. Department of Health and Human Services' <i>Physical Activity Guidelines for Americans</i> In tandem with the recommendations above, Americans of all ages—children, adolescents, adults, and older adults—should meet the <i>Physical Activity Guidelines for Americans</i> to help promote health and reduce the risk of chronic disease. Americans should aim to achieve and maintain a healthy body weight. The relationship between diet and physical activity contributes to calorie balance and managing body weight.

Source: Adapted from the *Dietary Guidelines for Americans, 2020–2025*. Washington, DC: U.S. Department of Agriculture and U.S. Department of Health and Human Services; 2020. Available at https://www.dietaryguidelines.gov/sites/default/files/2020-12/Dietary_Guidelines_for_Americans_2020-2025.pdf.

the general approach should focus on whole foods, eating patterns, and appropriate calorie balance, rather than on specific micronutrients such as electrolytes or selected vitamins. It should be remembered that most patients have difficulty understanding nutritional labels on packaged foods, with the attendant demands on numeracy and health literacy.

Dietary guidelines are published by the U.S. Department of Agriculture (USDA) and U.S. Department of Health and Human Services every 5 years, and these guidelines have undergone substantial evolution over time. The current U.S. Dietary Guidelines and Key Recommendations for 2020–2025 are summarized in Table 2-1 and emphasize the importance of healthy eating patterns for every stage of life, to avoid chronic diseases including obesity, diabetes, cancer, and cardiovascular disease. The core elements include eating patterns with nutrient-dense (rather than calorie-dense) whole foods and appropriate caloric intake to achieve and maintain healthy weight. The USDA guidelines focus on the concept of a healthy plate (rather than the prior food pyramid) for ease of counseling and adoption. Fifty percent of the plate should consist of vegetables and whole fruits, with remaining portions for whole grains and lean protein foods. When using fat for cooking, it should be done by sauteing in healthier oils (e.g., canola oil), and addition of judicious amounts of healthy raw oils (e.g., olive oil, nuts) to dishes is appropriate. Recommendations also focus on limitation of foods and beverages higher in added sugars, saturated fat, and sodium, and moderation or avoidance of alcohol intake.

The USDA guidelines focus on specific healthy eating patterns that adhere to these broad recommendations and are appropriate for ~97% of the general population. They identify a “Healthy U.S.-Style Dietary Pattern” that adheres closely to the evidence-based Dietary Approaches to Stop Hypertension (DASH) eating pattern but is customizable for different cultural or personal preferences. Alternative patterns, which vary more in emphasis than in content, include a “Healthy Mediterranean-Style Dietary Pattern” and a “Healthy Vegetarian Dietary Pattern.”

AGE- AND SEX-SPECIFIC RECOMMENDATIONS Current dietary framework recommendations are generally similar for all life stages from ages ≥ 12 months, but recommended levels of caloric intake (and hence amounts of foods) differ by age, sex, and physical activity level. For example, recommended caloric intake ranges from 1000 calories/d for sedentary 2-year-old children to as high as 3200 calories/d for active 16- to 18-year-old young men. Recommended caloric intakes peak in late adolescence or early adulthood for men and women and gradually decrease over ensuing decades.

As with all lifestyle counseling aimed at behavior change, dietary approaches that partner with the patient and utilize motivational interviewing strategies and shared goals and commitments tend to work best, as described below (see “Approach to the Patient”).

Physical Activity Similar to the approach to counseling regarding healthy eating patterns, recommendations on participation in physical activity emphasize the point that any physical activity is better than none. A simple rule of thumb for patients is: “If you are doing nothing, do something; and if you are doing something, do more, every day.” The evidence base for physical activity indicates that the marginal benefits from physical activity are greatest in advancing from no activity to low levels of moderate activity. With increasing duration and intensity of activity, there is a continued curvilinear increase in health benefits, but the marginal gains for each additional minute of moderate-to-vigorous activity slowly diminish. Thus, for adults, the recommended amount of physical activity is 150 min of moderate-intensity or 75 min of vigorous-intensity aerobic activity per week, performed in episodes of at least 5 min, and preferably spread throughout the week, plus participation in muscle-strengthening activity at least 2 days per week. Additional health benefits can be realized by engaging in physical activity beyond this amount.

In counseling patients regarding physical activity, it is important to note that sedentary time (e.g., seated at work or at home in front of electronic screens) has adverse health consequences independent of the lack of physical activity during these episodes. Therefore, even modest

efforts like standing at the desk and doing gentle stretching for periods during the day may be beneficial. It is also important to emphasize that participating in a variety of aerobic activities (biking, swimming, walking, jogging, rowing, elliptical training, stair-climbing, etc.) can be beneficial and may help to avoid overuse injuries and boredom with the exercise regimen. Addition of resistance (muscle-strengthening) activities is also beneficial for health improvement. Emphasis should be placed on body-weight resistance or weights that allow more repetitions (e.g., 3 sets of 15–20 repetitions that can be performed comfortably, with a rest period in between) and on avoiding breath-holding and straining against a closed glottis.

SUDDEN CARDIAC DEATH RISK Patients may express concerns regarding the risk of sudden cardiac death during exercise. Whereas the risk of sudden death during exercise does increase directly with the amount of time spent exercising, this association is substantially mitigated by training effects. Thus, patients embarking on an exercise program should be encouraged to increase the duration of aerobic exercise gradually as tolerated, aiming for episodes of at least 30 min 5 times a week as an ideal. Once a comfortable duration is reached, incorporating interval training periods of more intensive activity interspersed during the exercise can provide greater fitness gains.

EXTREME ENDURANCE ACTIVITIES As with other forms of exercise, extreme endurance activities such as triathlons and marathons should be undertaken only with appropriate and graded training. Such activities tend to take a greater toll on the musculoskeletal system over time than less extreme activities, and they are also associated with measurable damage to the myocardium and greater risks for other organ damage. Athletes participating in endurance activities routinely have elevations in cardiac troponin (a specific circulating marker of myocardial cell damage and death) at the end of the race, although elevations are lower in those who are well trained. Patients and clinicians should consider the patient’s overall health, specific limitations, potential for injury, and ability to train in decision-making regarding participation in endurance events.

AGE-SPECIFIC RECOMMENDATIONS The U.S. Department of Health and Human Services’ *Physical Activity Guidelines for Americans*, second edition (2018) (Table 2-2), recommend that preschool-aged children (aged 3–5 years) should be physically active throughout the day in a variety of activity types to enhance growth and development. Children and adolescents aged 6–17 years should participate in ≥ 60 min of physical activity daily, most of which should be moderate- or vigorous-intensity aerobic activity, including vigorous, muscle-strengthening, and bone-strengthening activities at least 3 days a week each. As noted above, adults aged 18–64 years are recommended to pursue at least 150 min of moderate-intensity or 75 min of vigorous-intensity aerobic activity per week (or equivalent combinations), with at least 2 days of muscle-strengthening activities. Adults aged ≥ 65 years should follow the adult guidelines or be as active as possible as abilities and conditions allow. For older adults, special emphasis is also placed on multicomponent physical activity that includes balance training as well as aerobic and muscle-strengthening activities.

Sleep Hygiene Sleeping between 7 and 9 h per night appears to be optimal for health in adults aged ≥ 18 years. Sleeping < 7 h is associated with adverse outcomes, including obesity, diabetes, elevated blood pressure, cardiovascular disease, depression, and all-cause mortality, as well as physiologic disturbances such as impaired immune function, increased pain sensitivity, and impaired cognitive performance. Conversely, achieving appropriate levels of sleep is associated with more success in weight loss, better blood pressure control among patients with hypertension, and improved mental health and performance. Regular sleep more than 9 h per night is appropriate for children and adolescents or individuals recovering from sleep deprivation or illness, but for most individuals, the effects on health are uncertain.

Patients often express concerns about the quantity and quality of their sleep. With aging, both aspects of sleep tend to decline, even without overt sleep disorders. Documentation of sleep using a sleep log may assist in understanding different types of insomnia and sleep disorders.

TABLE 2-2 Recommendations from *Physical Activity Guidelines for Americans, 2nd Edition (2018)*

AGE	RECOMMENDATIONS
3–5 years	- Preschool-aged children (ages 3 through 5 years) should be physically active throughout the day to enhance growth and development. - Adult caregivers of preschool-aged children should encourage active play that includes a variety of activity types.
6–17 years	- It is important to provide young people opportunities and encouragement to participate in physical activities that are appropriate for their age, that are enjoyable, and that offer variety. - Children and adolescents ages 6 through 17 years should do 60 min (1 h) or more of moderate-to-vigorous physical activity daily: - Aerobic: Most of the 60 min or more per day should be either moderate- or vigorous-intensity aerobic physical activity and should include vigorous-intensity physical activity on at least 3 days a week. - Muscle-strengthening: As part of their 60 min or more of daily physical activity, children and adolescents should include muscle-strengthening physical activity on at least 3 days a week. - Bone-strengthening: As part of their 60 min or more of daily physical activity, children and adolescents should include bone-strengthening physical activity on at least 3 days a week.
18–64 years	- Adults should move more and sit less throughout the day. Some physical activity is better than none. Adults who sit less and do any amount of moderate-to-vigorous physical activity gain some health benefits. - For substantial health benefits, adults should do at least 150 min (2 h and 30 min) to 300 min (5 h) a week of moderate-intensity or 75 min (1 h and 15 min) to 150 min (2 h and 30 min) a week of vigorous-intensity aerobic physical activity, or an equivalent combination of moderate- and vigorous-intensity aerobic activity. Preferably, aerobic activity should be spread throughout the week. - Additional health benefits are gained by engaging in physical activity beyond the equivalent of 300 min (5 h) of moderate-intensity physical activity a week. - Adults should also do muscle-strengthening activities of moderate or greater intensity and that involve all major muscle groups on 2 or more days a week, as these activities provide additional health benefits.
≥65 years	- The key guidelines for adults also apply to older adults. In addition, the following key guidelines are just for older adults: - As part of their weekly physical activity, older adults should do multicomponent physical activity that includes balance training as well as aerobic and muscle-strengthening activities. - Older adults should determine their level of effort for physical activity relative to their level of fitness. - Older adults with chronic conditions should understand whether and how their conditions affect their ability to do regular physical activity safely. - When older adults cannot do 150 min of moderate-intensity aerobic activity a week because of chronic conditions, they should be as physically active as their abilities and conditions allow.

Moderate-intensity physical activity: Aerobic activity that increases a person's heart rate and breathing to some extent. On a scale relative to a person's capacity, moderate-intensity activity is usually a 5 or 6 on a 0 to 10 scale. Brisk walking, dancing, swimming, or bicycling on a level terrain are examples. Vigorous-intensity physical activity: Aerobic activity that greatly increases a person's heart rate and breathing. On a scale relative to a person's capacity, vigorous-intensity activity is usually a 7 or 8 on a 0 to 10 scale. Jogging, singles tennis, swimming continuous laps, or bicycling uphill are examples. Muscle-strengthening activity: Physical activity, including exercise that increases skeletal muscle strength, power, endurance, and mass. It includes strength training, resistance training, and muscular strength and endurance exercises. Bone-strengthening activity: Physical activity that produces an impact or tension force on bones, which promotes bone growth and strength. Running, jumping rope, and lifting weights are examples.

Source: Adapted from U.S. Department of Health and Human Services. *Physical Activity Guidelines for Americans, 2nd edition*. Washington, DC: U.S. Department of Health and Human Services; 2018. Available at https://health.gov/sites/default/files/2019-09/Physical_Activity_Guidelines_2nd_edition.pdf.

Encouraging daily activity to promote fatigue, avoidance of eating and drinking alcohol too close to bedtime, and regular daily sleep habits may help patients achieve better sleep. Regular use of sedative medications should generally be discouraged given the high potential for dependence, addiction, and altered sleep quality.

DISORDERS OF SLEEP The prevalence of sleep-related breathing disorders, including obstructive sleep apnea (OSA), is high and increasing in our aging and increasingly overweight population. A recent systematic review suggested that the prevalence of clinically important OSA in the general adult population may be between 9 and 38%, with higher rates in men versus women, older versus younger adults, and those with higher versus lower body mass index (BMI). In the United States, ~40 million adults, and worldwide, nearly a billion adults, are affected by OSA. Patients with persistent complaints of poor sleep quality or excessive daytime somnolence or with witnessed apneic spells may benefit from screening for sleep disorders prior to consideration of a formal sleep study. A number of clinical tools have been developed to screen for sleep apnea, including the Epworth Sleepiness Scale, the STOP (snoring, tiredness, observed apnea, high blood pressure) Questionnaire, and the STOP-Bang Questionnaire (STOP plus assessment of BMI, age, neck circumference, and gender), among others. The U.S. Preventive Services Task Force found that current evidence is insufficient to assess the balance of benefits and harms of screening for OSA in asymptomatic adults owing to a lack of validation data in primary care settings. Nonetheless, the high prevalence and significant health consequences of sleep apnea suggest that clinicians should be alert for its potential presence, particularly in patients who are obese with symptoms of excessive daytime somnolence or witnessed apnea episodes. Other sleep disorders, such as restless leg syndrome, may be identified with simple history.

Weight Management Overweight and obesity are prevalent in epidemic proportions in the United States and other industrialized nations (**Chaps. 413 and 414**). Since 1985, the prevalence of obesity in the United States has increased from ~10 to ~35%, and the prevalence of overweight is now ~40%. Overweight and obesity disproportionately affect individuals in lower socioeconomic strata and in many underrepresented populations, including individuals who identify as Black, Latino, and American Indian. In all race/ethnic groups, both overweight and obesity are associated with adverse health consequences, including diabetes, certain cancers, cardiovascular diseases, and degenerative joint disease. Eating disorders such as anorexia and bulimia are much less common but pose major health consequences for affected patients and should be suspected particularly in younger women with history of rapid weight shifts, electrolyte disturbances, or underweight status.

Weight loss is one of the most difficult preventive interventions to achieve and sustain over time. However, several key factors can assist the patient and clinician, and early referral to a dietician can be very helpful. The first therapeutic goal is to aim for weight stabilization. Many of the risks of overweight and obesity are driven more strongly by continued weight gain, rather than overweight/obese status per se. Working with the patient to find initial strategies for weight maintenance can be a successful initial step with success for many patients. For those who can progress to considering weight loss, it is critical to help the patient understand that there is no standard solution. Experimentation and documentation are key. Tools to assist patients can include food and weight logs, activity logs, and smart phone apps. Some patients respond best to structured approaches such as intermittent fasting regimens or commercial dietary programs where meals are

provided. Any of these approaches can be tried with or without social group supports.

The key construct for weight loss is, of course, negative calorie balance. This is achieved through a combination of reduced caloric intake and increased physical activity. Patients may already understand, from prior weight loss attempts, what combination works best for them to achieve this. Some patients find that they cannot lose weight without increasing their exercise. For many, reduction of caloric intake is most efficient. Encouraging the patient to find what works for them is most important. The same principle holds for dietary content. Well-done feeding studies indicate that weight loss is dependent far more on the reduction of caloric intake than on the relative composition of fat, protein, and carbohydrate in the diet. There may be other medical reasons to choose one approach over another, but if not, encouraging the patient to pick one approach and document the results is an important start. Newer pharmacologic agents (e.g., GLP1 receptor agonists) appear to be remarkably successful in helping patients achieve substantial weight loss (up to 20% of baseline body weight), with proven short-term benefits in cardiovascular risk reduction, but issues of access, cost, and long-term safety remain. Once weight loss is achieved, increase in activity is often required for its successful maintenance.

Tobacco Cessation (see Chap. 465) Escaping nicotine dependence is another major, but critical, challenge to prevention and wellness efforts. The addictive effects of nicotine have been well documented, with effects that can last for years after successful cessation. Assessing a patient's past history of cessation attempts and current readiness for change are key first steps in forging a successful approach. Frequent follow-up and reinforcement, as well as use of nicotine replacement therapy and other cessation-promoting medications, are additional critical elements. Recidivism is the rule, and patients should expect to resume smoking and attempt again as they journey to tobacco cessation. Electronic cigarettes have some evidence for benefit in adult cessation of combustible tobacco, but this approach may transfer and worsen nicotine addiction, and the potential for use by adolescents and young adults who are not smokers represents a major public health threat for a new generation of nicotine addiction, with unknown health consequences as a result of the high doses of nicotine delivered to developing organs, including the brain. Vaping of other substances, often in association with flavoring compounds, has also been associated with pulmonary and cardiovascular damage and should be actively discouraged.

■ VACCINATION (CHAP. 129)

One of the major advances in public health that has contributed to increases in health and longevity worldwide is the development of safe and effective vaccinations against endemic and epidemic infectious diseases. Patients should be counseled regarding age-appropriate vaccinations for their children and for themselves. Some individuals may be reluctant to receive a vaccination; in these cases, listening to the patient's concerns is important, followed by explanation of the benefits to the individual, their family, and their community and review of the low risk for potential harms. It is true to say that no current vaccines are ever worse than the disease they prevent, although side effects may occur rarely. Thorough knowledge of the data on side effect rates and of efficacy will aid the clinician in helping the patient make a fully informed decision.

■ MENTAL HEALTH AND ADDICTION

Assessment for depression and cognitive impairment is important to address when patients exhibit symptoms or they or their family members express concerns. Both of these common conditions play a major role in reducing quality of life and are high on patients' lists of concerns, even if not clearly expressed. Screening tools for depression are reviewed in Chap. 463. Cognitive function decline with aging or comorbid illness, including depression, should be anticipated. Assessment tools such as the General Practitioner Assessment of Cognition or the Mini-Cog test are widely available and effective rapid assessment tools.

Alcohol and Opioids (see Chaps. 464 and 467) Alcohol dependence and abuse are common and underdiagnosed. Rapid

screening tools have proven efficacy for identifying patients with alcohol problems. In a systematic review, the CAGE (cut down, annoyed, guilty, eye opener) questionnaire was most effective at identifying alcohol abuse and dependence, with reasonable sensitivity and high specificity. The present opioid epidemic in the United States presents a new and substantial public health challenge given the high potential for dependency and abuse of these drugs. Rapid screening tools are available to assist clinicians in screening for opioid dependence.

■ ACCIDENTS AND SUICIDE

Regular assessment of patient safety through simple questions about seat belt use, domestic violence, and gun safety in the home continues to be an important part of health promotion and wellness. Longstanding recommendations for assessment of suicidal ideation among patients with depression or a history of suicide attempts also continue to be relevant.

APPROACH TO THE PATIENT

In the context of a clinical visit focused on health assessment, health promotion, and prevention, the basic skills of history-taking are of paramount importance. Much of the evaluation, counseling, and management that focus on health promotion and prevention also require engagement and buy-in from the patient in order to assist with recognition of contributing behaviors and to promote adherence to therapeutic plans. Therefore, in addition to standard history-taking, additional skills such as motivational interviewing and eliciting patient commitments and contracting may prove of significant value. The availability of additional tools to assist with screening, monitoring, and chronic management, both online and through wearable devices and mobile health technologies, is rapidly expanding, with uncertain implications for the future. Major research gaps exist in our understanding of how best to employ these newer technologies to improve health outcomes. Concepts of behavioral economics are being explored to better understand the psychology of decision-making and incentives as a means to improve lifestyle choices and adherence to treatment plans (Chap. 494).

The limited time available to clinicians and patients during a wellness visit or periodic health examination (not driven by specific patient issues) makes it important to prioritize assessment and counseling for factors that affect longevity, health span, and quality of life over approaches that may have low yield, such as the annual comprehensive physical examination in an asymptomatic patient. Setting clear expectations for the content of a wellness visit may be a first step, and scheduling follow-up visits for findings or to continue indicated counseling are important steps to achieving better health outcomes.

■ FURTHER READING

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